

Predictors of Earnings Risk with Machine Learning

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Abstract

This paper looks at the predictors of lifetime earnings risk under a Restricted Income Profile (RIP) model using traditional and machine learning methods such as lasso and SHAP values. The paper builds on the work of Drewianka and Oberg (2025) which uses a moment condition approach derive a parameter that captures permanent income risk. The paper finds that education and age are important in explaining lifetime earnings risk. The paper also finds that macroeconomic variables such as probability of recession and real GDP growth are important and along with state controls may further imply a role of government policy. Finally, the paper finds that occupation controls are important while industry controls do not appear to play a strong role.

Keywords: machine learning, restricted income profile, earnings instability, risk

JEL Codes: D8, J0, D3

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- 1 Introduction
- 2 Theoretical Framework
- 3 Data
- 4 Empirical Strategy
- 5 Results
 - 5.1 Permanent Shocks

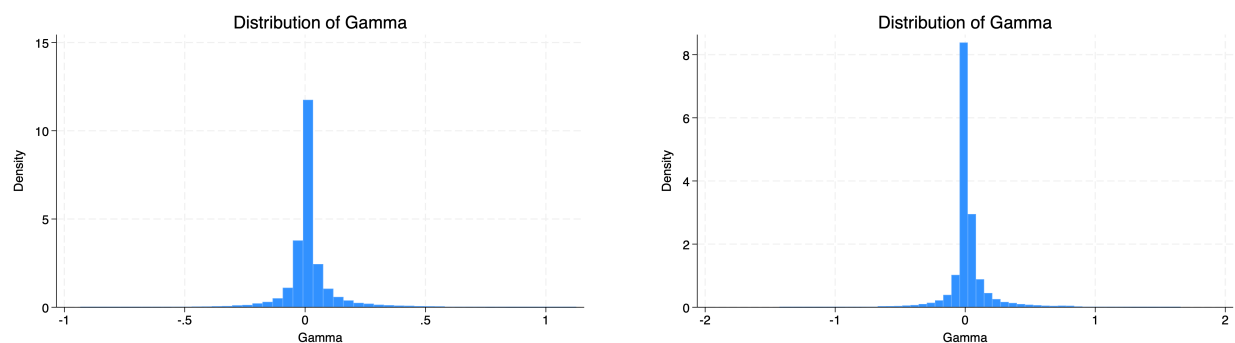


Figure 1: Distribution of Gamma: Hourly earnings (fhwage, left) and Annual earnings (fearn, right)

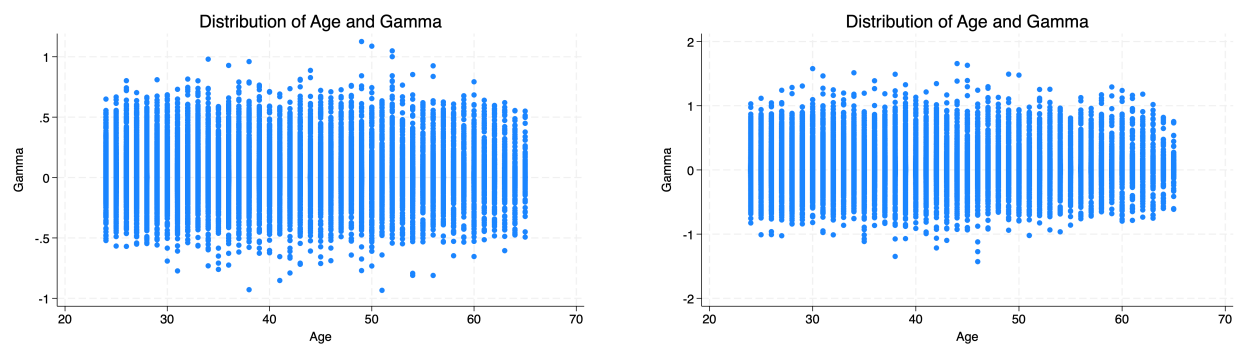


Figure 2: Scatterplot of Age vs. Gamma: Hourly earnings (fhwage, left) and Annual earnings (fearn, right)

Table 1: Gamma Regressions: F-Test Results (Hourly Earnings, fh wage)

	No Occ/Ind (1)	No Occ (2)	No Ind (3)	All Controls (4)
Census Division FE	1.92** (0.0439)	1.89** (0.0484)	1.75* (0.0730)	1.64* (0.0990)
Year FE	1.15 (0.2539)	1.17 (0.2285)	1.19 (0.2171)	1.16 (0.2451)
Race FE	1.85 (0.1004)	1.51 (0.1840)	1.26 (0.2765)	1.29 (0.2664)
Cohort FE	4.36*** (0.0045)	4.33*** (0.0046)	4.61*** (0.0032)	4.54*** (0.0035)
Occupation FE	—	—	2.27*** (0.0000)	2.00*** (0.0000)
Industry FE	—	2.61*** (0.0000)	—	1.96*** (0.0009)
Education FE	3.45*** (0.0079)	3.84*** (0.0040)	2.14* (0.0737)	2.59** (0.0346)
Age Bin FE	3.44*** (0.0041)	3.04*** (0.0095)	3.37*** (0.0048)	3.15*** (0.0077)
Tenure FE	0.75 (0.4720)	0.40 (0.6728)	0.62 (0.5397)	0.52 (0.5927)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 2: Gamma Regressions: F-Test Results (Annual Earnings, fearn)

	No Occ/Ind (1)	No Occ (2)	No Ind (3)	All Controls (4)
Census Division FE	2.66*** (0.0044)	2.55*** (0.0064)	2.72*** (0.0036)	2.49*** (0.0076)
Year FE	1.69*** (0.0090)	2.39*** (0.0000)	2.27*** (0.0001)	2.37*** (0.0000)
Race FE	2.23** (0.0485)	2.07* (0.0659)	2.24** (0.0476)	2.28** (0.0438)
Cohort FE	4.46*** (0.0039)	5.11*** (0.0016)	4.78*** (0.0025)	4.99*** (0.0018)
Occupation FE	—	—	2.78*** (0.0000)	1.95*** (0.0000)
Industry FE	—	4.18*** (0.0000)	—	2.19*** (0.0001)
Education FE	3.45*** (0.0080)	3.17** (0.0131)	1.80 (0.1259)	2.04* (0.0854)
Age Bin FE	4.21*** (0.0008)	3.72*** (0.0023)	3.89*** (0.0016)	3.59*** (0.0030)
Tenure FE	2.20 (0.1105)	0.37 (0.6904)	0.53 (0.5883)	0.38 (0.6809)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

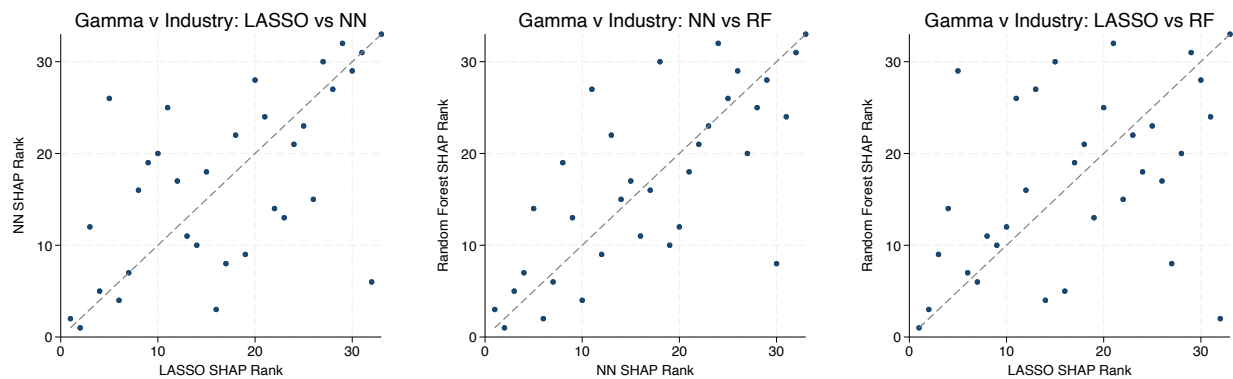


Figure 3: Gamma Industry SHAP Rank Comparisons (Hourly Earnings, flwage)

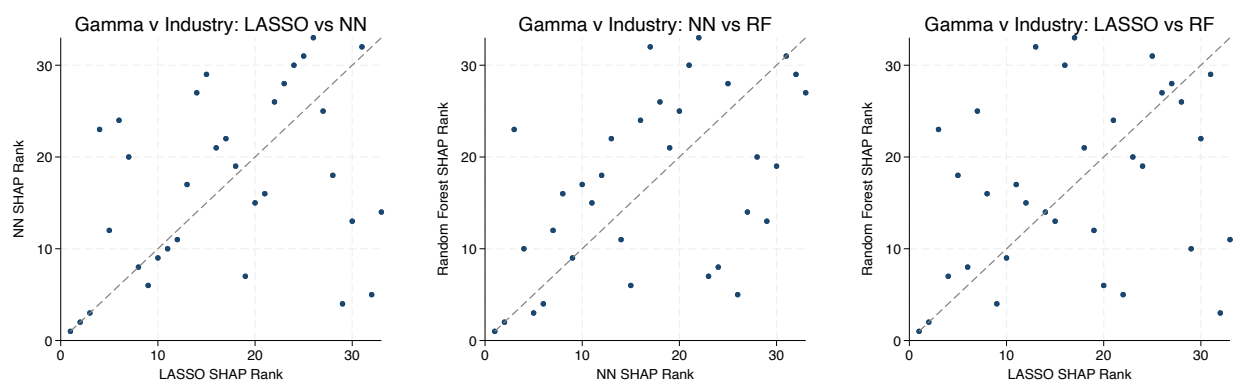


Figure 4: Gamma Industry SHAP Rank Comparisons (Annual Earnings, fearn)

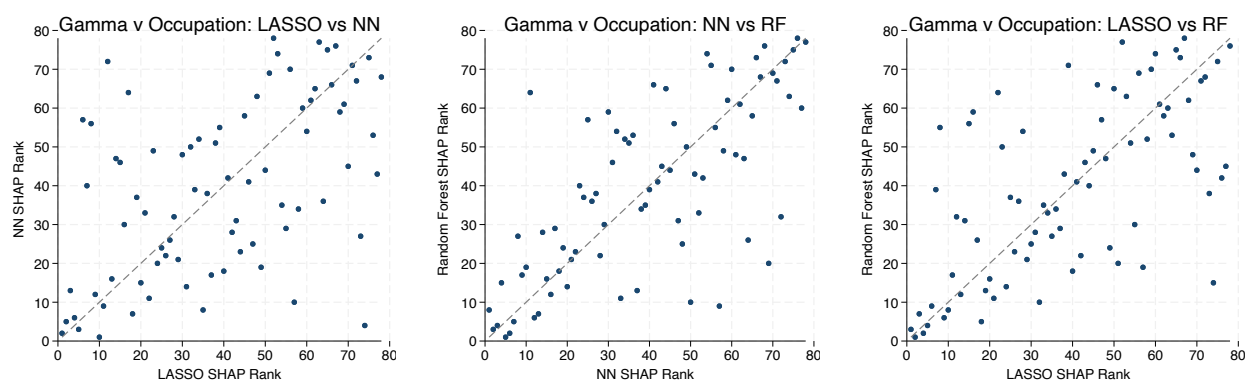


Figure 5: Gamma Occupation SHAP Rank Comparisons (Hourly Earnings, flwage)

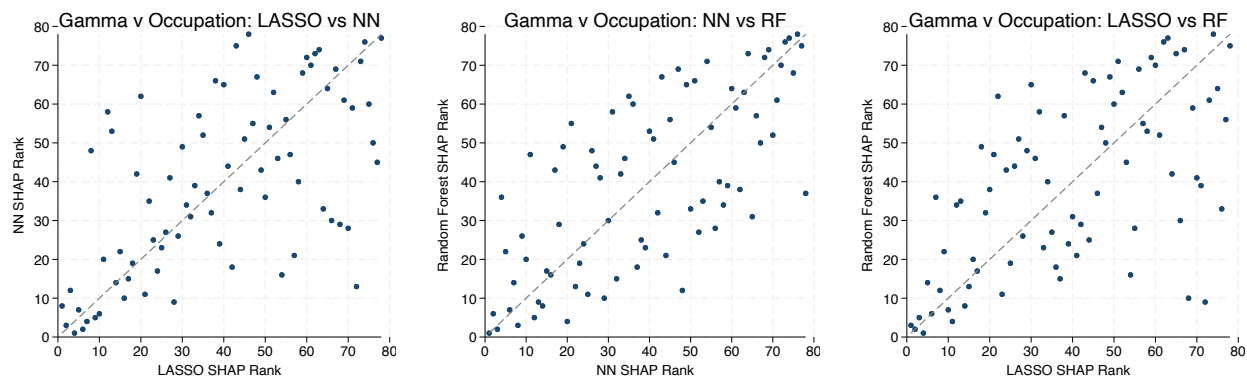


Figure 6: Gamma Occupation SHAP Rank Comparisons (Annual Earnings, fearn)

Table 3: Gamma Regressions: OLS Results (Coefficients x 100), Hourly Earnings (flhwage)

	(1)	(2)	(3)	(4)	(5)
	Gamma (flhwage)	Gamma (flhwage)	Gamma (flhwage)	Gamma (flhwage)	Gamma (flhwage)
High School Graduate	0.0130 (0.114)	0.0556 (0.120)	0.109 (0.121)	0.0847 (0.121)	0.108 (0.122)
Some College	0.249* (0.143)	0.293* (0.150)	0.347** (0.152)	0.293* (0.155)	0.325** (0.156)
Bachelor's Degree	0.390*** (0.141)	0.435*** (0.150)	0.514*** (0.155)	0.395** (0.162)	0.451*** (0.163)
Bachelors+	0.223 (0.174)	0.288 (0.185)	0.369* (0.195)	0.221 (0.207)	0.261 (0.208)
22-29	0.00116 (0.00135)	0.00524*** (0.00197)	0.00486** (0.00197)	0.00546*** (0.00197)	0.00525*** (0.00197)
30-37	0.000891 (0.00120)	0.00396** (0.00155)	0.00373** (0.00155)	0.00423*** (0.00155)	0.00404*** (0.00155)
38-45	-0.00115 (0.00125)	0.000517 (0.00134)	0.000393 (0.00134)	0.000744 (0.00134)	0.000623 (0.00134)
54-61	-0.0000366 (0.00156)	-0.000872 (0.00160)	-0.00106 (0.00160)	-0.00120 (0.00160)	-0.00130 (0.00160)
62-69	0.00815** (0.00328)	0.00658** (0.00334)	0.00581* (0.00335)	0.00559* (0.00335)	0.00512 (0.00335)
Probability of Recession	-0.00200 (0.00225)	1.775 (4.805)	2.907 (4.851)	3.373 (4.835)	3.604 (4.861)
5-year moving average of AEP	0.00275 (0.00201)	0.00296 (0.00218)	0.000733 (0.00228)	0.00341 (0.00233)	0.00238 (0.00239)
Out of Labor Force	-0.478 (0.364)	-0.491 (0.365)	-0.624* (0.369)	-0.604 (0.369)	-0.644* (0.369)
2-5	-0.000246 (0.000985)	-0.000314 (0.00111)	0.000284 (0.00112)	0.0000930 (0.00112)	0.000320 (0.00112)
6+	-0.00125 (0.000964)	-0.00130 (0.00109)	-0.000707 (0.00111)	-0.00105 (0.00111)	-0.000821 (0.00111)
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	0.000	0.001	0.002	0.003	0.004
N	82112	82112	82112	82112	82112

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Gamma Regressions: OLS Results (Coefficients x 100), Annual Earnings (fearn)

	(1)	(2)	(3)	(4)	(5)
	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)
High School Graduate	0.0677 (0.169)	0.129 (0.177)	0.207 (0.178)	0.191 (0.179)	0.229 (0.180)
Some College	0.492** (0.211)	0.574*** (0.221)	0.646*** (0.224)	0.549** (0.229)	0.592** (0.230)
Bachelor's Degree	0.435** (0.209)	0.531** (0.221)	0.577** (0.229)	0.412* (0.239)	0.486** (0.241)
Bachelors+	0.474* (0.258)	0.665** (0.274)	0.676** (0.289)	0.553* (0.305)	0.602* (0.308)
22-29	0.000594 (0.00198)	0.00347 (0.00290)	0.00240 (0.00290)	0.00321 (0.00291)	0.00293 (0.00291)
30-37	0.00144 (0.00176)	0.00400* (0.00228)	0.00358 (0.00228)	0.00420* (0.00229)	0.00398* (0.00229)
38-45	-0.0000964 (0.00183)	0.00165 (0.00196)	0.00146 (0.00196)	0.00185 (0.00196)	0.00174 (0.00197)
54-61	0.00429* (0.00241)	0.00358 (0.00247)	0.00296 (0.00247)	0.00289 (0.00247)	0.00257 (0.00247)
62-69	0.0251*** (0.00569)	0.0240*** (0.00580)	0.0229*** (0.00580)	0.0230*** (0.00580)	0.0222*** (0.00581)
Probability of Recession	-0.000859 (0.00331)	-4.014 (7.178)	1.837 (7.235)	1.910 (7.214)	2.992 (7.251)
5-year moving average of AEP	0.0240*** (0.00293)	0.0262*** (0.00318)	0.0210*** (0.00334)	0.0242*** (0.00341)	0.0232*** (0.00350)
Out of Labor Force	-1.035* (0.589)	-1.148* (0.589)	-1.787*** (0.595)	-1.709*** (0.595)	-1.814*** (0.595)
2-5	-0.00162 (0.00145)	-0.00248 (0.00162)	-0.000518 (0.00164)	-0.000927 (0.00164)	-0.000517 (0.00164)
6+	-0.00247* (0.00142)	-0.00315** (0.00160)	-0.00140 (0.00163)	-0.00168 (0.00164)	-0.00143 (0.00164)
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	0.001	0.003	0.004	0.005	0.006
N	80661	80661	80661	80661	80661

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Gamma Regressions: Stepwise Selection Results (Coefficients x 100), Hourly Earnings (flhwage)

	(1)	(2)	(3)	(4)	(5)
	Gamma (flhwage)	Gamma (flhwage)	Gamma (flhwage)	Gamma (flhwage)	Gamma (flhwage)
5-year moving average of AEP				0.00401*	
				(0.00198)	
Constant	0.0139***	0.0129***	0.00789***	0.0131***	0.0146***
	(0.00124)	(0.00161)	(0.00207)	(0.00342)	(0.00334)
Census Division FE		✓	✓		
Year FE					
Race FE					
Cohort FE					
Occupation FE				✓	✓
Industry FE			✓		✓
Education FE	✓	✓	✓		
Age Bin FE	✓	✓			
Tenure FE					
Census Division FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
Education FE Available	Yes	Yes	Yes	Yes	Yes
Age Bin FE Available	Yes	Yes	Yes	Yes	Yes
Tenure FE Available	Yes	Yes	Yes	Yes	Yes
R-squared	0.000	0.001	0.001	0.002	0.003
N	82112	82112	82112	82112	82112

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Gamma Regressions: Stepwise Selection Results (Coefficients x 100), Annual Earnings (fearn)

	(1)	(2)	(3)	(4)	(5)
	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)	Gamma (fearn)
5-year moving average of AEP	0.0246*** (0.00285)	0.0263*** (0.00298)	0.0210*** (0.00318)	0.0230*** (0.00321)	0.0216*** (0.00330)
Out of Labor Force			-1.721** (0.594)	-1.606** (0.593)	-1.714** (0.594)
Constant	0.0108*** (0.00228)	0.00821* (0.00327)	0.00368 (0.00400)	-0.0132 (0.00935)	-0.0132 (0.00936)
Census Division FE		✓	✓	✓	✓
Year FE			✓	✓	✓
Race FE				✓	✓
Cohort FE		✓	✓	✓	✓
Occupation FE				✓	✓
Industry FE			✓		✓
Education FE	✓	✓	✓		
Age Bin FE	✓	✓	✓	✓	✓
Tenure FE					
Census Division FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
Education FE Available	Yes	Yes	Yes	Yes	Yes
Age Bin FE Available	Yes	Yes	Yes	Yes	Yes
Tenure FE Available	Yes	Yes	Yes	Yes	Yes
R-squared	0.001	0.002	0.004	0.005	0.006
N	80661	80661	80661	80661	80661

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7: Mean Predicted Gamma by Demographic Group, Hourly Earnings (flwage)

Group	Category	Actual	NN	RF	LASSO	N
Education	Less than HS	0.0146	0.0216	0.0144	0.0149	13,247
Education	High School Graduate	0.0146	0.0220	0.0145	0.0145	34,816
Education	Some College	0.0167	0.0227	0.0156	0.0167	11,987
Education	Bachelor's Degree	0.0178	0.0225	0.0165	0.0177	14,697
Education	Bachelors+	0.0162	0.0225	0.0163	0.0161	6,764
Race	White	0.0161	0.0223	0.0154	0.0161	54,090
Race	Black	0.0143	0.0217	0.0140	0.0143	20,065
Race	Native American	0.0166	0.0228	0.0156	0.0161	1,820
Race	Asian or Pacific Islander	0.0101	0.0224	0.0171	0.0113	697
Race	Latino	0.0168	0.0225	0.0160	0.0166	4,371
Race	Other or Unknown	0.0066	0.0212	0.0145	0.0083	468
Cohort	born pre-1944	0.0159	0.0222	0.0152	0.0158	19,850
Cohort	born 1944-1952	0.0145	0.0215	0.0138	0.0145	21,430
Cohort	born 1953-1960	0.0153	0.0222	0.0150	0.0153	19,459
Cohort	born post-1960	0.0168	0.0228	0.0166	0.0169	20,772
Age Bin	22-29	0.0168	0.0220	0.0159	0.0169	16,169
Age Bin	30-37	0.0162	0.0220	0.0156	0.0159	24,113
Age Bin	38-45	0.0140	0.0221	0.0140	0.0144	18,415
Age Bin	46-53 (ref)	0.0150	0.0224	0.0141	0.0142	13,654
Age Bin	54-61	0.0149	0.0226	0.0154	0.0164	7,922
Age Bin	62-69	0.0236	0.0230	0.0234	0.0218	1,238
Tenure Bin	0-1 (ref)	0.0161	0.0221	0.0150	0.0159	39,662
Tenure Bin	2-5	0.0159	0.0221	0.0151	0.0155	18,671
Tenure Bin	6+	0.0144	0.0223	0.0154	0.0152	23,178

Table 8: Mean Predicted Gamma by Demographic Group, Annual Earnings (fearn)

Group	Category	Actual	NN	RF	LASSO	N
Education	Less than HS	0.0257	0.0417	0.0255	0.0257	12,793
Education	High School Graduate	0.0235	0.0381	0.0240	0.0235	34,355
Education	Some College	0.0263	0.0409	0.0260	0.0262	11,816
Education	Bachelor's Degree	0.0233	0.0371	0.0242	0.0233	14,515
Education	Bachelors+	0.0230	0.0360	0.0240	0.0229	6,581
Race	White	0.0241	0.0376	0.0242	0.0241	52,885
Race	Black	0.0252	0.0424	0.0259	0.0252	19,929
Race	Native American	0.0197	0.0374	0.0244	0.0202	1,778
Race	Asian or Pacific Islander	0.0212	0.0361	0.0232	0.0223	679
Race	Latino	0.0247	0.0378	0.0253	0.0245	4,321
Race	Other or Unknown	0.0051	0.0344	0.0147	0.0068	468
Cohort	born pre-1944	0.0248	0.0387	0.0246	0.0247	18,365
Cohort	born 1944-1952	0.0218	0.0363	0.0238	0.0218	21,252
Cohort	born 1953-1960	0.0241	0.0404	0.0250	0.0241	19,521
Cohort	born post-1960	0.0262	0.0398	0.0250	0.0261	20,922
Age Bin	22-29	0.0267	0.0421	0.0277	0.0277	16,476
Age Bin	30-37	0.0245	0.0390	0.0249	0.0241	24,324
Age Bin	38-45	0.0216	0.0375	0.0221	0.0209	18,437
Age Bin	46-53 (ref)	0.0215	0.0371	0.0213	0.0215	13,306
Age Bin	54-61	0.0261	0.0368	0.0262	0.0280	6,671
Age Bin	62-69	0.0491	0.0367	0.0494	0.0401	846
Tenure Bin	0-1 (ref)	0.0261	0.0404	0.0269	0.0256	39,275
Tenure Bin	2-5	0.0235	0.0380	0.0236	0.0238	18,426
Tenure Bin	6+	0.0213	0.0366	0.0213	0.0220	22,359

Table 9: Mean Predicted Gamma by Education and Age Bin, Hourly Earnings (flwage)

Actual						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0151	0.0162	0.0162	0.0116	0.0113	0.0342
High School Graduate	0.0149	0.0148	0.0133	0.0152	0.0136	0.0236
Some College	0.0184	0.0179	0.0133	0.0190	0.0101	0.0406
Bachelor's Degree	0.0218	0.0168	0.0159	0.0171	0.0226	0.0014
Bachelors+	0.0279	0.0195	0.0111	0.0130	0.0200	0.0203

Neural Network						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0214	0.0214	0.0215	0.0218	0.0221	0.0226
High School Graduate	0.0218	0.0218	0.0220	0.0223	0.0225	0.0229
Some College	0.0226	0.0225	0.0226	0.0229	0.0232	0.0234
Bachelor's Degree	0.0223	0.0223	0.0225	0.0227	0.0231	0.0233
Bachelors+	0.0227	0.0224	0.0222	0.0225	0.0229	0.0232

Random Forest						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0146	0.0149	0.0142	0.0134	0.0135	0.0258
High School Graduate	0.0150	0.0148	0.0135	0.0138	0.0142	0.0234
Some College	0.0162	0.0159	0.0142	0.0151	0.0154	0.0265
Bachelor's Degree	0.0183	0.0167	0.0152	0.0151	0.0186	0.0152
Bachelors+	0.0263	0.0182	0.0131	0.0134	0.0177	0.0263

LASSO						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0163	0.0155	0.0140	0.0133	0.0151	0.0212
High School Graduate	0.0159	0.0146	0.0132	0.0132	0.0151	0.0203
Some College	0.0185	0.0168	0.0151	0.0152	0.0175	0.0228
Bachelor's Degree	0.0191	0.0179	0.0167	0.0163	0.0188	0.0243
Bachelors+	0.0190	0.0170	0.0143	0.0149	0.0177	0.0219

Table 10: Mean Predicted Gamma by Education and Age Bin, Annual Earnings (fearn)

Actual						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0289	0.0287	0.0256	0.0237	0.0189	0.0247
High School Graduate	0.0262	0.0253	0.0197	0.0193	0.0232	0.0496
Some College	0.0262	0.0248	0.0252	0.0277	0.0283	0.0689
Bachelor's Degree	0.0264	0.0200	0.0202	0.0228	0.0384	0.0336
Bachelors+	0.0290	0.0239	0.0206	0.0153	0.0271	0.0713

Neural Network						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0460	0.0424	0.0406	0.0397	0.0395	0.0417
High School Graduate	0.0412	0.0382	0.0366	0.0362	0.0355	0.0351
Some College	0.0449	0.0411	0.0392	0.0386	0.0383	0.0392
Bachelor's Degree	0.0393	0.0375	0.0364	0.0359	0.0361	0.0347
Bachelors+	0.0389	0.0372	0.0355	0.0354	0.0349	0.0345

Random Forest						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0298	0.0274	0.0237	0.0223	0.0236	0.0337
High School Graduate	0.0269	0.0244	0.0216	0.0207	0.0247	0.0407
Some College	0.0281	0.0258	0.0245	0.0233	0.0284	0.0501
Bachelor's Degree	0.0275	0.0237	0.0208	0.0215	0.0302	0.0537
Bachelors+	0.0304	0.0247	0.0202	0.0191	0.0273	0.0699

LASSO						
Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.0292	0.0260	0.0228	0.0228	0.0291	0.0455
High School Graduate	0.0271	0.0234	0.0201	0.0207	0.0264	0.0380
Some College	0.0301	0.0259	0.0227	0.0237	0.0305	0.0435
Bachelor's Degree	0.0263	0.0233	0.0204	0.0207	0.0280	0.0380
Bachelors+	0.0280	0.0234	0.0195	0.0209	0.0276	0.0381

5.2 Transitory Shocks

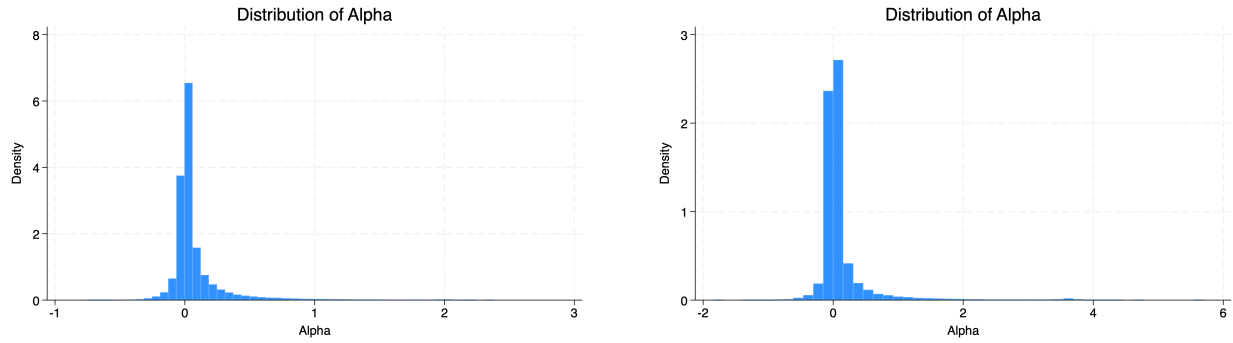


Figure 7: Distribution of Alpha: Hourly earnings (fhwage, left) and Annual earnings (fearn, right)

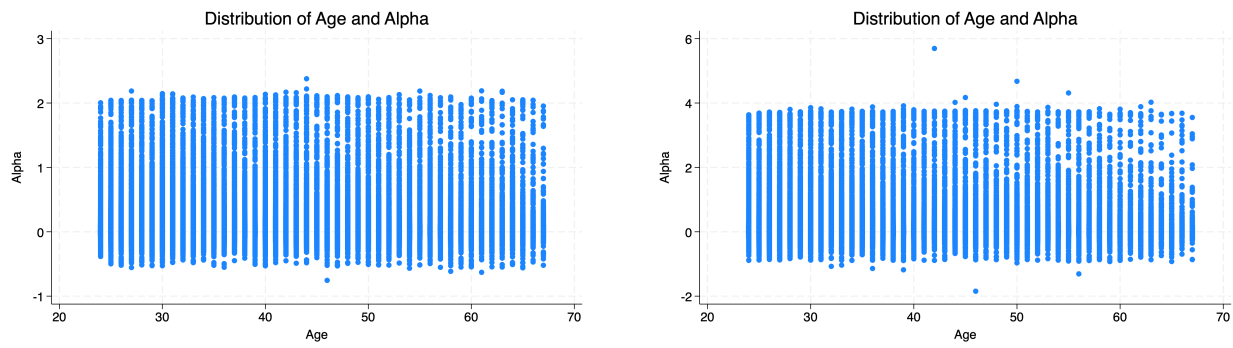


Figure 8: Scatterplot of Age vs. Alpha: Hourly earnings (fhwage, left) and Annual earnings (fearn, right)

Table 11: Alpha Regressions: F-Test Results (Hourly Earnings, flwage)

	No Occ/Ind (1)	No Occ (2)	No Ind (3)	All Controls (4)
Census Division FE	14.86*** (0.0000)	8.37*** (0.0000)	9.50*** (0.0000)	7.83*** (0.0000)
Year FE	8.86*** (0.0000)	11.02*** (0.0000)	10.85*** (0.0000)	11.11*** (0.0000)
Race FE	7.08*** (0.0000)	8.84*** (0.0000)	9.52*** (0.0000)	10.18*** (0.0000)
Cohort FE	3.21** (0.0220)	4.87*** (0.0022)	4.13*** (0.0062)	4.83*** (0.0023)
Occupation FE	—	—	17.12*** (0.0000)	7.00*** (0.0000)
Industry FE	—	34.54*** (0.0000)	—	10.21*** (0.0000)
Education FE	9.38*** (0.0000)	12.75*** (0.0000)	5.41*** (0.0002)	6.99*** (0.0000)
Age Bin FE	13.90*** (0.0000)	11.82*** (0.0000)	12.15*** (0.0000)	11.16*** (0.0000)
Tenure FE	42.01*** (0.0000)	24.17*** (0.0000)	25.83*** (0.0000)	21.84*** (0.0000)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 12: Alpha Regressions: F-Test Results (Annual Earnings, fearn)

	No Occ/Ind (1)	No Occ (2)	No Ind (3)	All Controls (4)
Census Division FE	12.64*** (0.0000)	9.19*** (0.0000)	10.45*** (0.0000)	9.49*** (0.0000)
Year FE	5.09*** (0.0000)	10.36*** (0.0000)	9.86*** (0.0000)	10.44*** (0.0000)
Race FE	7.81*** (0.0000)	9.54*** (0.0000)	11.36*** (0.0000)	11.64*** (0.0000)
Cohort FE	2.65** (0.0468)	3.85*** (0.0091)	3.59** (0.0131)	3.82*** (0.0094)
Occupation FE	—	—	14.21*** (0.0000)	4.72*** (0.0000)
Industry FE	—	29.83*** (0.0000)	—	7.01*** (0.0000)
Education FE	16.53*** (0.0000)	22.15*** (0.0000)	7.32*** (0.0000)	9.26*** (0.0000)
Age Bin FE	3.71*** (0.0023)	4.35*** (0.0006)	4.18*** (0.0009)	4.13*** (0.0009)
Tenure FE	89.55*** (0.0000)	41.12*** (0.0000)	42.54*** (0.0000)	36.80*** (0.0000)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

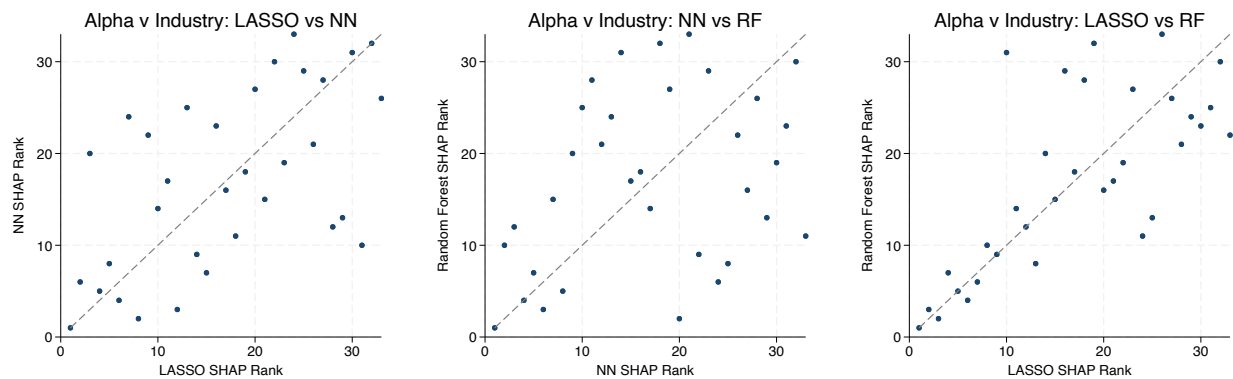


Figure 9: Alpha Industry SHAP Rank Comparisons (Hourly Earnings, flwage)

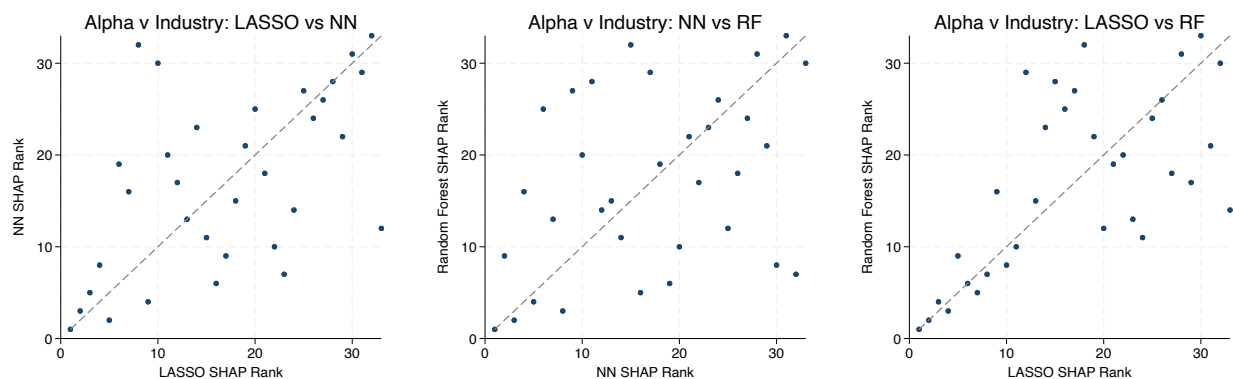


Figure 10: Alpha Industry SHAP Rank Comparisons (Annual Earnings, fearn)

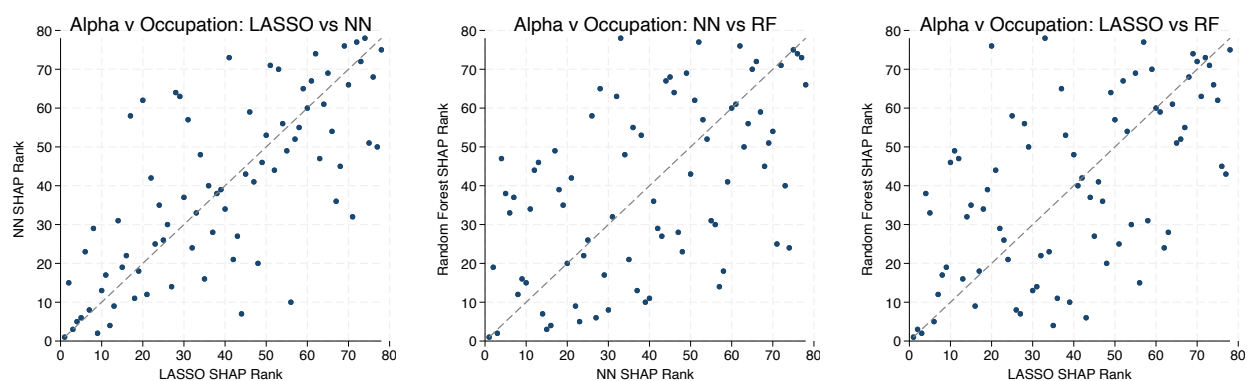


Figure 11: Alpha Occupation SHAP Rank Comparisons (Hourly Earnings, flwage)

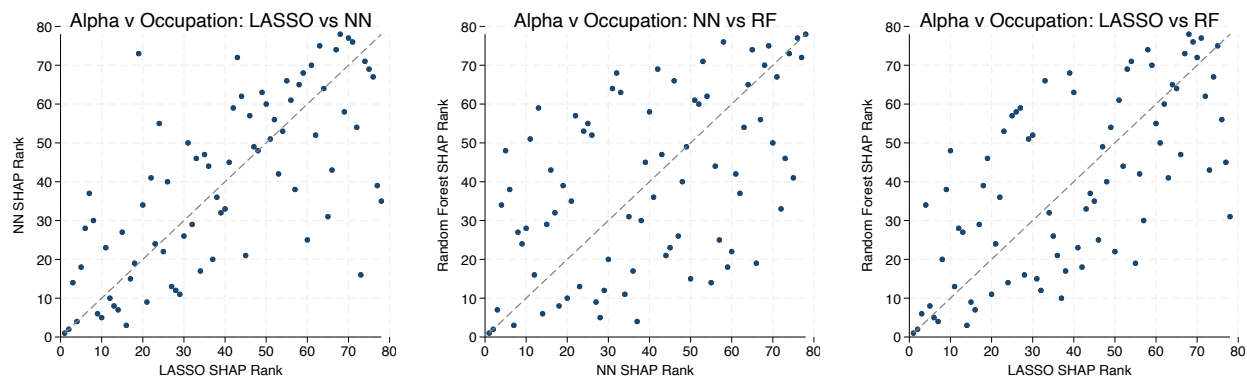


Figure 12: Alpha Occupation SHAP Rank Comparisons (Annual Earnings, fearn)

Table 13: Alpha Regressions: OLS Results (Coefficients x 100), Hourly Earnings (flwage)

	(1)	(2)	(3)	(4)	(5)
	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)
High School Graduate	0.836*** (0.248)	0.557** (0.259)	1.060*** (0.260)	1.018*** (0.261)	1.140*** (0.262)
Some College	1.341*** (0.313)	1.019*** (0.329)	1.538*** (0.331)	1.297*** (0.338)	1.440*** (0.339)
Bachelor's Degree	1.836*** (0.311)	1.618*** (0.329)	2.107*** (0.339)	1.374*** (0.353)	1.633*** (0.356)
Bachelors+	2.127*** (0.383)	2.013*** (0.407)	2.635*** (0.426)	1.651*** (0.451)	1.889*** (0.454)
22-29	-0.0215*** (0.00301)	-0.0123*** (0.00433)	-0.0130*** (0.00432)	-0.0118*** (0.00432)	-0.0121*** (0.00432)
30-37	-0.0119*** (0.00267)	-0.00519 (0.00342)	-0.00497 (0.00341)	-0.00416 (0.00341)	-0.00450 (0.00341)
38-45	-0.00324 (0.00278)	-0.000335 (0.00296)	-0.000341 (0.00295)	0.0000983 (0.00295)	-0.000125 (0.00295)
54-61	0.00145 (0.00335)	0.00196 (0.00344)	0.000119 (0.00342)	0.000435 (0.00342)	0.0000342 (0.00342)
62-69	0.0437*** (0.00563)	0.0443*** (0.00579)	0.0388*** (0.00577)	0.0402*** (0.00577)	0.0379*** (0.00577)
Probability of Recession	0.0104** (0.00505)	0.470 (0.912)	2.314** (0.917)	2.542*** (0.913)	2.724*** (0.916)
5-year moving average of AEP	0.131*** (0.00435)	0.135*** (0.00470)	0.112*** (0.00491)	0.124*** (0.00500)	0.119*** (0.00513)
Out of Labor Force	7.676*** (0.719)	7.554*** (0.720)	5.768*** (0.727)	5.797*** (0.725)	5.605*** (0.726)
2-5	-0.0223*** (0.00222)	-0.0201*** (0.00251)	-0.0147*** (0.00253)	-0.0150*** (0.00252)	-0.0138*** (0.00252)
6+	-0.0256*** (0.00213)	-0.0187*** (0.00246)	-0.0151*** (0.00249)	-0.0158*** (0.00249)	-0.0145*** (0.00250)
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	.0172843	.0217232	.0322897	.0343087	.0374175
N	101315	101315	101315	101315	101315

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Alpha Regressions: OLS Results (Coefficients x 100), Annual Earnings (fearn)

	(1)	(2)	(3)	(4)	(5)
	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)
High School Graduate	1.124*** (0.433)	0.582 (0.453)	1.236*** (0.454)	1.061** (0.456)	1.268*** (0.457)
Some College	1.164** (0.546)	0.622 (0.573)	1.399** (0.578)	0.764 (0.589)	1.039* (0.591)
Bachelor's Degree	3.516*** (0.543)	3.099*** (0.574)	4.161*** (0.590)	2.574*** (0.616)	3.062*** (0.620)
Bachelors+	4.635*** (0.669)	4.307*** (0.711)	5.759*** (0.745)	3.669*** (0.788)	4.148*** (0.794)
22-29	-0.0275*** (0.00520)	-0.0134* (0.00756)	-0.0184** (0.00754)	-0.0159** (0.00754)	-0.0172** (0.00754)
30-37	-0.0157*** (0.00463)	-0.00624 (0.00595)	-0.00694 (0.00593)	-0.00532 (0.00594)	-0.00606 (0.00594)
38-45	-0.000839 (0.00481)	0.00424 (0.00514)	0.00369 (0.00512)	0.00478 (0.00512)	0.00434 (0.00512)
54-61	0.00191 (0.00600)	-0.00159 (0.00614)	-0.00487 (0.00612)	-0.00489 (0.00612)	-0.00515 (0.00612)
62-69	0.0426*** (0.0113)	0.0370*** (0.0116)	0.0310*** (0.0116)	0.0315*** (0.0116)	0.0288** (0.0116)
Probability of Recession	0.0705*** (0.00879)	3.539** (1.612)	7.878*** (1.618)	7.783*** (1.613)	8.243*** (1.619)
5-year moving average of AEP	0.366*** (0.00751)	0.377*** (0.00813)	0.344*** (0.00852)	0.363*** (0.00867)	0.363*** (0.00890)
Out of Labor Force	16.78*** (1.402)	16.17*** (1.403)	11.03*** (1.414)	11.30*** (1.411)	10.69*** (1.412)
2-5	-0.0528*** (0.00385)	-0.0536*** (0.00435)	-0.0375*** (0.00438)	-0.0383*** (0.00438)	-0.0358*** (0.00438)
6+	-0.0461*** (0.00371)	-0.0434*** (0.00427)	-0.0285*** (0.00433)	-0.0287*** (0.00434)	-0.0265*** (0.00435)
Census Division FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	.0366571	.0396923	.0488128	.0501382	.0522766
N	99639	99639	99639	99639	99639

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 15: Alpha Regressions: Stepwise Selection Results (Coefficients x 100), Hourly Earnings (flwage)

	(1)	(2)	(3)	(4)	(5)
	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)	Alpha (flwage)
Probability of Recession	0.0104* (0.00505)			0.0222* (0.0113)	
5-year moving average of AEP	0.131*** (0.00435)	0.132*** (0.00469)	0.108*** (0.00490)	0.121*** (0.00499)	0.116*** (0.00512)
Out of Labor Force	7.676*** (0.719)	7.668*** (0.720)	5.894*** (0.728)	5.931*** (0.726)	5.725*** (0.726)
Constant	0.0433*** (0.00376)	0.0465*** (0.0125)	0.0171 (0.0129)	0.0389** (0.0146)	0.0371* (0.0146)
Census Division FE		✓	✓	✓	✓
Year FE		✓	✓	✓	✓
Race FE		✓	✓	✓	✓
Cohort FE		✓	✓	✓	✓
Occupation FE				✓	✓
Industry FE			✓		✓
Education FE	✓	✓	✓	✓	✓
Age Bin FE	✓	✓	✓	✓	✓
Tenure FE	✓	✓	✓	✓	✓
Census Division FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
Education FE Available	Yes	Yes	Yes	Yes	Yes
Age Bin FE Available	Yes	Yes	Yes	Yes	Yes
Tenure FE Available	Yes	Yes	Yes	Yes	Yes
R-squared	0.017	0.021	0.031	0.033	0.036
N	101315	101315	101315	101315	101315

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 16: Alpha Regressions: Stepwise Selection Results (Coefficients x 100), Annual Earnings (fearn)

	(1)	(2)	(3)	(4)	(5)
	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)	Alpha (fearn)
Probability of Recession	0.0705*** (0.00879)	0.0752*** (0.0197)	0.0740*** (0.0196)	0.0752*** (0.0196)	0.0739*** (0.0196)
5-year moving average of AEP	0.366*** (0.00751)	0.374*** (0.00811)	0.340*** (0.00850)	0.360*** (0.00866)	0.360*** (0.00889)
Out of Labor Force	16.78*** (1.402)	16.27*** (1.403)	11.14*** (1.414)	11.44*** (1.411)	10.81*** (1.412)
Constant	-0.0118 (0.00654)	-0.0201 (0.0217)	-0.0732** (0.0224)	-0.0697** (0.0253)	-0.0745** (0.0253)
Census Division FE		✓	✓	✓	✓
Year FE		✓	✓	✓	✓
Race FE		✓	✓	✓	✓
Cohort FE		✓	✓	✓	✓
Occupation FE				✓	✓
Industry FE			✓		✓
Education FE	✓	✓	✓	✓	✓
Age Bin FE	✓	✓	✓	✓	✓
Tenure FE	✓	✓	✓	✓	✓
Census Division FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
Education FE Available	Yes	Yes	Yes	Yes	Yes
Age Bin FE Available	Yes	Yes	Yes	Yes	Yes
Tenure FE Available	Yes	Yes	Yes	Yes	Yes
R-squared	0.037	0.039	0.048	0.050	0.052
N	99639	99639	99639	99639	99639

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 17: Mean Predicted Alpha by Demographic Group, Hourly Earnings (fhwage)

Group	Category	Actual	NN	RF	LASSO	N
Education	Less than HS	0.0993	0.1321	0.0945	0.0992	17,294
Education	High School Graduate	0.0916	0.1250	0.0894	0.0916	43,010
Education	Some College	0.0884	0.1237	0.0877	0.0885	14,528
Education	Bachelor's Degree	0.0792	0.1185	0.0814	0.0793	17,489
Education	Bachelors+	0.0820	0.1207	0.0835	0.0821	8,220
Race	White	0.0859	0.1217	0.0856	0.0859	65,542
Race	Black	0.1009	0.1308	0.0936	0.1008	25,891
Race	Native American	0.0816	0.1278	0.0904	0.0811	2,193
Race	Asian or Pacific Islander	0.0692	0.1200	0.0888	0.0704	857
Race	Latino	0.0863	0.1273	0.0902	0.0865	5,478
Race	Other or Unknown	0.0850	0.1317	0.0996	0.0868	580
Cohort	born pre-1944	0.0866	0.1242	0.0890	0.0865	24,778
Cohort	born 1944-1952	0.0857	0.1236	0.0862	0.0857	25,694
Cohort	born 1953-1960	0.0904	0.1230	0.0869	0.0904	23,984
Cohort	born post-1960	0.0953	0.1272	0.0903	0.0953	26,085
Age Bin	22-29	0.0926	0.1265	0.0914	0.0931	19,269
Age Bin	30-37	0.0854	0.1227	0.0844	0.0859	28,867
Age Bin	38-45	0.0861	0.1233	0.0834	0.0837	22,310
Age Bin	46-53 (ref)	0.0865	0.1243	0.0848	0.0857	16,605
Age Bin	54-61	0.0903	0.1259	0.0924	0.0983	10,724
Age Bin	62-69	0.1548	0.1368	0.1456	0.1383	2,766
Tenure Bin	0-1 (ref)	0.1070	0.1293	0.1046	0.1016	50,885
Tenure Bin	2-5	0.0747	0.1220	0.0735	0.0819	21,501
Tenure Bin	6+	0.0693	0.1179	0.0696	0.0735	28,155

Table 18: Mean Predicted Alpha by Demographic Group, Annual Earnings (fearn)

Group	Category	Actual	NN	RF	LASSO	N
Education	Less than HS	0.1653	0.1360	0.1487	0.1650	16,727
Education	High School Graduate	0.1366	0.1230	0.1370	0.1367	42,477
Education	Some College	0.1120	0.1075	0.1218	0.1121	14,362
Education	Bachelor's Degree	0.0964	0.0891	0.1011	0.0964	17,278
Education	Bachelors+	0.0967	0.0960	0.1063	0.0968	8,031
Race	White	0.1128	0.1051	0.1136	0.1128	64,168
Race	Black	0.1673	0.1416	0.1625	0.1673	25,709
Race	Native American	0.1279	0.1105	0.1343	0.1274	2,156
Race	Asian or Pacific Islander	0.0967	0.0986	0.1190	0.0979	842
Race	Latino	0.1221	0.1086	0.1329	0.1223	5,422
Race	Other or Unknown	0.1038	0.1014	0.1406	0.1055	578
Cohort	born pre-1944	0.1106	0.1054	0.1181	0.1105	23,055
Cohort	born 1944-1952	0.1208	0.1098	0.1235	0.1209	25,494
Cohort	born 1953-1960	0.1401	0.1236	0.1383	0.1401	24,041
Cohort	born post-1960	0.1378	0.1199	0.1317	0.1378	26,285
Age Bin	22-29	0.1572	0.1314	0.1601	0.1555	19,603
Age Bin	30-37	0.1232	0.1156	0.1240	0.1262	29,060
Age Bin	38-45	0.1163	0.1080	0.1125	0.1118	22,347
Age Bin	46-53 (ref)	0.1115	0.1055	0.1098	0.1111	16,316
Age Bin	54-61	0.1203	0.1086	0.1263	0.1302	9,622
Age Bin	62-69	0.1988	0.1235	0.2053	0.1763	1,927
Tenure Bin	0-1 (ref)	0.1685	0.1305	0.1681	0.1554	50,330
Tenure Bin	2-5	0.0899	0.1037	0.0881	0.1130	21,250
Tenure Bin	6+	0.0817	0.0947	0.0852	0.0879	27,295

Table 19: Mean Predicted Alpha by Education and Age Bin, Hourly Earnings (flwage)

Actual							Neural Network						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.1234	0.0998	0.0894	0.0841	0.0879	0.1686	Less than HS	0.1344	0.1310	0.1307	0.1294	0.1318	0.1500
High School Graduate	0.0934	0.0929	0.0900	0.0867	0.0797	0.1537	High School Graduate	0.1265	0.1235	0.1243	0.1244	0.1255	0.1375
Some College	0.0851	0.0801	0.0861	0.0896	0.1142	0.1494	Some College	0.1261	0.1221	0.1225	0.1236	0.1257	0.1335
Bachelor's Degree	0.0704	0.0690	0.0795	0.0876	0.0872	0.1687	Bachelor's Degree	0.1193	0.1169	0.1181	0.1194	0.1202	0.1247
Bachelors+	0.0563	0.0669	0.0774	0.0850	0.1060	0.1216	Bachelors+	0.1213	0.1188	0.1186	0.1223	0.1225	0.1287

Random Forest							LASSO						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.1035	0.0940	0.0862	0.0846	0.0912	0.1631	Less than HS	0.1037	0.0966	0.0910	0.0901	0.1049	0.1586
High School Graduate	0.0916	0.0876	0.0874	0.0851	0.0900	0.1391	High School Graduate	0.0942	0.0889	0.0873	0.0877	0.0991	0.1422
Some College	0.0900	0.0820	0.0810	0.0880	0.1033	0.1530	Some College	0.0925	0.0849	0.0828	0.0858	0.0997	0.1364
Bachelor's Degree	0.0809	0.0759	0.0775	0.0831	0.0895	0.1427	Bachelor's Degree	0.0799	0.0759	0.0758	0.0783	0.0893	0.1152
Bachelors+	0.0808	0.0753	0.0776	0.0834	0.0945	0.1259	Bachelors+	0.0850	0.0764	0.0739	0.0815	0.0927	0.1225

Table 20: Mean Predicted Alpha by Education and Age Bin, Annual Earnings (fearn)

Actual							Neural Network						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.2439	0.1642	0.1470	0.1298	0.1375	0.2245	Less than HS	0.1589	0.1405	0.1284	0.1223	0.1268	0.1647
High School Graduate	0.1658	0.1395	0.1243	0.1117	0.1052	0.1760	High School Graduate	0.1367	0.1243	0.1163	0.1113	0.1127	0.1308
Some College	0.1193	0.1058	0.1073	0.1036	0.1235	0.2234	Some College	0.1195	0.1084	0.1019	0.0992	0.1038	0.1154
Bachelor's Degree	0.0827	0.0858	0.0991	0.1022	0.1139	0.2058	Bachelor's Degree	0.0973	0.0898	0.0866	0.0846	0.0870	0.0899
Bachelors+	0.0760	0.0823	0.0825	0.0974	0.1319	0.1812	Bachelors+	0.1097	0.0999	0.0921	0.0926	0.0938	0.1056

Random Forest							LASSO						
Education	Age Bin						Education	Age Bin					
	22-29	30-37	38-45	46-53	54-61	62-69		22-29	30-37	38-45	46-53	54-61	62-69
Less than HS	0.2029	0.1515	0.1252	0.1214	0.1365	0.2341	Less than HS	0.2003	0.1691	0.1480	0.1409	0.1629	0.2377
High School Graduate	0.1671	0.1338	0.1222	0.1127	0.1270	0.2085	High School Graduate	0.1597	0.1363	0.1217	0.1172	0.1337	0.1845
Some College	0.1416	0.1163	0.1077	0.1095	0.1355	0.2232	Some College	0.1355	0.1111	0.0982	0.0953	0.1172	0.1630
Bachelor's Degree	0.1112	0.0960	0.0943	0.0951	0.1109	0.1812	Bachelor's Degree	0.1165	0.0957	0.0863	0.0843	0.1024	0.1358
Bachelors+	0.1290	0.1027	0.0946	0.0998	0.1150	0.1797	Bachelors+	0.1264	0.0987	0.0834	0.0906	0.1045	0.1452

6 Conclusion

References

- [1] Abowd, J. M. and Card, D. (1989). On the Covariance Structure of Earnings and Hours Changes. *Econometrica*, 57(2), 411–445.
- [2] Baker, M. (1997). Growth-Rate Heterogeneity and the Covariance Structure of Life-Cycle Earnings. *Journal of Labor Economics*, 15(2), 338–375.
- [3] Consumer Price Index for All Urban Consumers (2017). Data set, series CUUR0000SA0. Washington, D.C.: U.S. Bureau of Labor Statistics.

- [4] Drewianka, S. (2010). Cross-Sectional Variation in Individuals' Earnings Instability. *Review of Income and Wealth*, 56(2), 291–326.
- [5] Drewianka, Scott and Oberg, Phillip. (2025). Earnings Risk and Heterogeneous Expected Earnings Profiles. *Quantitative Economics*. Forthcoming.
- [6] Drewianka, S. and Oberg, P. (2019). Estimating Heterogeneity in Lifetime Earnings Risk. Unpublished manuscript, University of Wisconsin–Milwaukee and Illinois Wesleyan University. Available at <http://dx.doi.org/10.2139/ssrn.3630477>.
- [7] Guvenen, F. (2009). An Empirical Investigation of Labor Income Processes. *Review of Economic Dynamics*, 12, 58–79.
- [8] Guvenen, F. (2007). Learning Your Earning: Are Labor Income Shocks Really Very Persistent? *American Economic Review*, 97(3), 687–712.
- [9] Hill, M. S. (1992). *The Panel Study of Income Dynamics: A User's Guide*, Volume 2. Newbury Park, CA: Sage Publications.
- [10] Hryshko, D. (2012). Labor Income Profiles Are Not Heterogeneous: Evidence from Income Growth Rates. *Quantitative Economics*, 3, 177–209.
- [11] Lundberg, S. M., and Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30.
- [12] MaCurdy, T. E. (1982). The Use of Time Series Processes to Model the Error Structure of Earnings in a Longitudinal Data Analysis. *Journal of Econometrics*, 18(1), 83–114.
- [13] Meghir, C. and Pistaferri, L. (2004). Income Variance Dynamics and Heterogeneity. *Econometrica*, 72(1), 1–32.
- [14] Panel Study of Income Dynamics (2023). Public use dataset. Ann Arbor, MI: Institute for Social Research, Survey Research Center, University of Michigan.

Average Log Earnings by Age, by Group

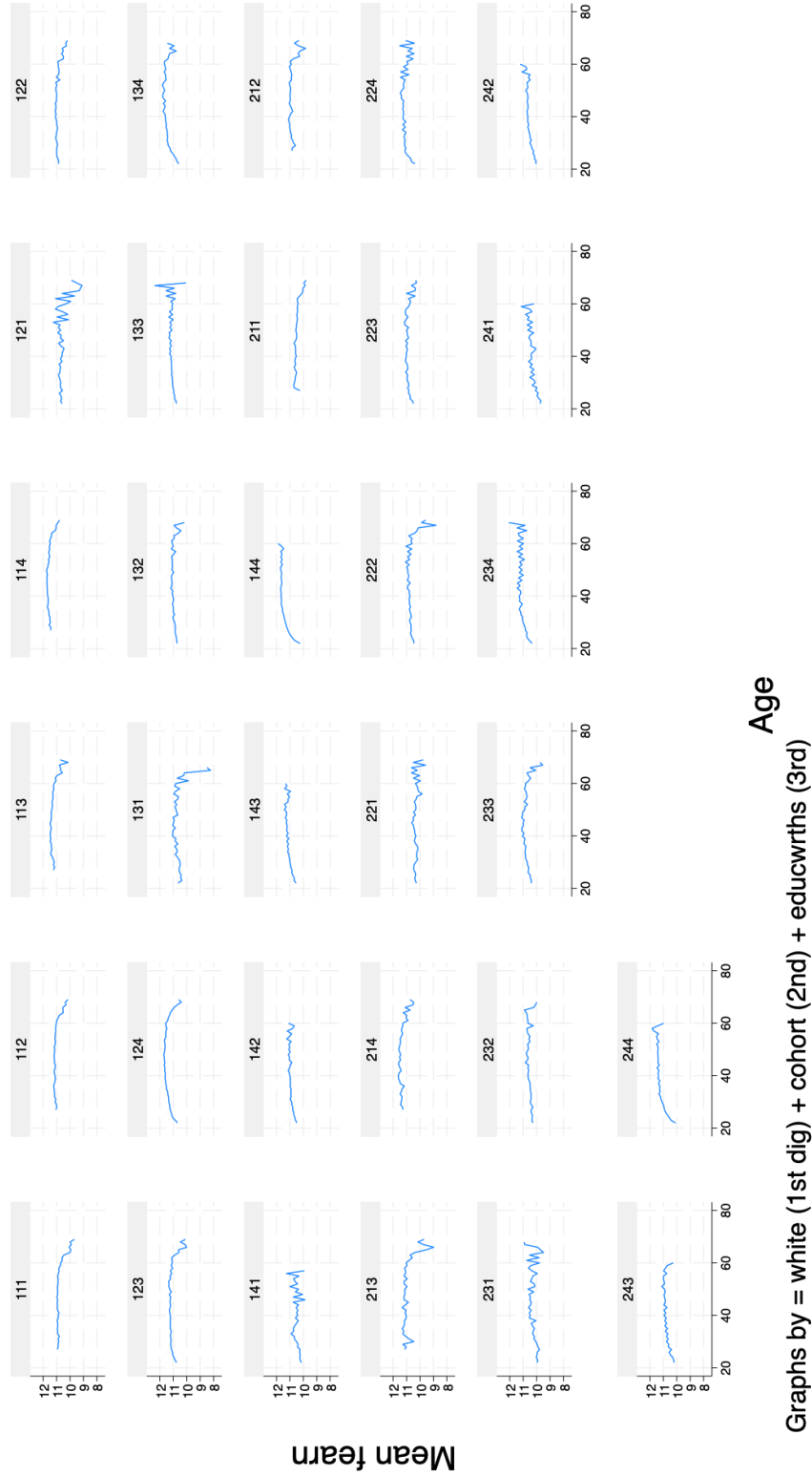


Figure 13: Mean Log Annual Earnings (fearn) by Age, by Group. Each panel is one group cell, where group = white/non-white (1st digit) $\times$ birth cohort (2nd digit) $\times$ education (3rd digit).

A Appendix

Table 21: Gamma Regressions: Industry Selection Results

Hourly Earnings (flwage)				Annual Earnings (fearn)			
Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Other Services	2	1	3	Not Applicable	1	1	1
Not Applicable	1	2	1	Other Services	2	2	2
Retail	16	3	5	Public Administration	3	3	23
Constr. Relate	6	4	7	Retail	29	4	10
Health Service	4	5	14	Agric.,Forestry	32	5	3
Agric.,Forestry	32	6	2	Legal Services	9	6	4
Construction	7	7	6	Mechanical Eng	19	7	12
Electrical Eng	17	8	19	Educ./Sport	8	8	16
Wood/Paper/Print	19	9	13	Other Trans.	10	9	9
Financial Inst	14	10	4	Wholesale	11	10	17
Energy/Water	13	11	27	Energy/Water	12	11	15
Public Administration	3	12	9	Health Service	5	12	18
Volunt./Church	23	13	22	Postal System	30	13	22
Mining	22	14	15	Electrical Eng	33	14	11
Service Indust	26	15	17	Construction	20	15	6
Legal Services	8	16	11	Iron/Steel	21	16	24
Educ./Sport	12	17	16	Clothing/Text	13	17	32
Clothing/Text	15	18	30	Chemicals	28	18	26
Wholesale	9	19	10	Food Industry	18	19	21
Mechanical Eng	10	20	12	Insurance	7	20	25
Social Security	24	21	18	Volunt./Church	16	21	30
Restaurants	18	22	21	Synthetics	17	22	33
Postal System	25	23	23	Constr. Relate	4	23	7
Earth/Clay/Stone	21	24	32	Wood/Paper/Print	6	24	8
Chemicals	11	25	26	Train System	27	25	28
Insurance	5	26	29	Service Indust	22	26	5
Iron/Steel	28	27	20	Financial Inst	14	27	14
Food Industry	20	28	25	Mining	23	28	20
Synthetics	30	29	28	Restaurants	15	29	13
Other Trans.	27	30	8	Earth/Clay/Stone	24	30	19
Fisheries	31	31	24	Social Security	25	31	31
Train System	29	32	31	Fisheries	31	32	29
Priv. Household	33	33	33	Priv. Household	26	33	27

Table 22: Gamma Regressions: Occupation Selection Results

Hourly Earnings (flwage)				Annual Earnings (fearn)			
Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	10	1	8	Not Applicable	4	1	1
Priv.Bus.Leadr	1	2	3	Transport.Oper	6	2	6
Vendor	5	3	4	Farm Manager	2	3	2
Machine Fitter	74	4	15	Architect/Engineer	7	4	36
Dr./Dentist/Vet	2	5	1	Electr. Enginr	9	5	22
Farm Manager	4	6	2	Insurance Rep.	10	6	7
Painter	18	7	5	Educator	5	7	14
Pipe Fitter	35	8	27	Priv.Bus.Leadr	1	8	3
Educator	11	9	17	Convey. Oper.	28	9	26
Mathematician	57	10	19	SecurityServic	16	10	20
Tool/Die Maker	22	11	64	Inspector	21	11	47
Insurance Rep.	9	12	6	Labor/Craftsmn	3	12	5
Legislator	3	13	7	Machine Fitter	72	13	9
Inspector	31	14	28	Painter	14	14	8
RelatMedicalJob	20	15	16	Mathematician	17	15	17
Lumbrman/Axman	13	16	12	Ofc.Worker Etc	54	16	16
Scientist	37	17	29	Accountant	24	17	43
BusinessManagr	40	18	18	Pipe Fitter	42	18	29
Soldier	49	19	24	Mailman	18	19	49
Bricklay/Carpt	24	20	14	Dr./Dentist/Vet	11	20	4
Office Manager	29	21	23	Soldier	57	21	55
Janitor	26	22	21	Tool/Die Maker	15	22	13
Bookkeepr/Cash	44	23	40	Vendor	25	23	19
Ofc.Worker Etc	25	24	37	BusinessManagr	39	24	24
Mailman	47	25	57	Janitor	23	25	11
SecurityServic	27	26	36	RelatMedicalJob	29	26	48
Life/Phys. Sci.	73	27	38	Domestic Help	26	27	44
Transport.Oper	42	28	22	Rest./StoreMgr	70	28	41
Administrator	55	29	30	Bricklay/Carpt	68	29	10
Chemist	16	30	59	Scientist	66	30	30
Buyer	43	31	46	Food Producer	32	31	58
Cook/Waiter	28	32	54	Sculptr/Paintr	37	32	15
Labor/Craftsmn	21	33	11	Life/Phys. Sci.	64	33	42
Tailor	58	34	52	Economist	31	34	46
Agriculturladm	54	35	51	Chemist	22	35	62
Miner	64	36	53	Stat.Mach.Oper	50	36	60
Aero/MarineEngr	19	37	13	Buyer	36	37	18
Stat.Mach.Oper	36	38	34	Printer Etc.	44	38	25
Rest./StoreMgr	33	39	35	Office Manager	33	39	23
Lawyer	7	40	39	Author	58	40	53
Foundry Worker	46	41	66	Lawyer	27	41	51
Printer Etc.	41	42	41	Cook/Waiter	19	42	32
Food Producer	77	43	45	ComputerOperat	49	43	67
Domestic Help	50	44	65	Lumbrman/Axman	41	44	21
Cleric	70	45	44	Cleric	77	45	56
Eng.Tech.Expert	15	46	56	Service Worker	53	46	45
Architect/Engineer	14	47	31	Foundry Worker	56	47	69
Hair Stylist	30	48	25	Bookkeepr/Cash	8	48	12
Author	23	49	50	Aero/MarineEngr	30	49	65
Service Worker	32	50	10	ChemicalWorker	76	50	33
Tel. Operator	38	51	43	Cabinet Maker	45	51	66
Prof. Athlete	34	52	33	Tailor	35	52	27
Economist	76	53	42	Dry-Cleaner	13	53	35
Cabinet Maker	60	54	74	Spinner/Weaver	51	54	71
ComputerOperat	39	55	71	Hair Stylist	47	55	54
Accountant	8	56	55	Music/Perform	55	56	28
Convey. Oper.	6	57	9	Eng.Tech.Expert	34	57	40
Sculptr/Paintr	45	58	49	Legislator	12	58	34
ChemicalWorker	68	59	62	Administrator	71	59	39
Dry-Cleaner	59	60	70	Stenographer	75	60	64
Fisher/Hunter	69	61	48	Miner	69	61	59
Salesperson	61	62	61	Tel. Operator	20	62	38
Transp. Attend	48	63	47	Agriculturladm	52	63	63
Music/Perform	17	64	26	Forestry Work	65	64	73
Tech.Salespers	62	65	58	Farm Hand	40	65	31
Forestry Work	66	66	73	Prof. Athlete	38	66	57
Shoemaker	72	67	68	Transp. Attend	48	67	50
Glazier	78	68	76	Shoemaker	59	68	72
Farm Hand	51	69	20	Glazier	67	69	74
Stenographer	56	70	69	Broadcaster	61	70	52
Broadcaster	71	71	67	Salesperson	73	71	61
Electr. Enginr	12	72	32	Jewelry Maker	60	72	70
Spinner/Weaver	75	73	72	Manufacturer	62	73	76
Jewelry Maker	53	74	63	Conductor	63	74	77
Manufacturer	65	75	75	Fisher/Hunter	43	75	68
HH Supervisor	67	76	78	Stone Cutter	74	76	78
Conductor	63	77	60	HH Supervisor	78	77	75
Stone Cutter	52	78	77	Tech.Salespers	46	78	37

Table 23: Alpha Regressions: Lasso Industry Selection Results

Hourly Earnings (flhwage)				Annual Earnings (fearn)			
Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	1	1	1	Not Applicable	1	1	1
Public Administration	8	2	10	Public Administration	5	2	9
Mechanical Eng	12	3	12	Other Services	2	3	2
Agric./Forestry	6	4	4	Educ./Sport	9	4	16
Retail	4	5	7	Agric./Forestry	3	5	4
Other Services	2	6	3	Energy/Water	16	6	25
Wood/Paper/Print	15	7	15	Mechanical Eng	23	7	13
Legal Services	5	8	5	Constr. Relate	4	8	3
Educ./Sport	14	9	20	Postal System	17	9	27
Food Industry	31	10	25	Iron/Steel	22	10	20
Iron/Steel	18	11	28	Electrical Eng	15	11	28
Postal System	28	12	21	Wood/Paper/Print	33	12	14
Electrical Eng	29	13	24	Health Service	13	13	15
Chemicals	10	14	31	Wholesale	24	14	11
Energy/Water	21	15	17	Clothing/Text	18	15	32
Health Service	17	16	18	Retail	7	16	5
Wholesale	11	17	14	Chemicals	12	17	29
Synthetics	19	18	32	Food Industry	21	18	19
Clothing/Text	23	19	27	Legal Services	6	19	6
Constr. Relate	3	20	2	Other Trans.	11	20	10
Train System	26	21	33	Synthetics	19	21	22
Other Trans.	9	22	9	Financial Inst	29	22	17
Insurance	16	23	29	Insurance	14	23	23
Construction	7	24	6	Mining	26	24	26
Service Indust	13	25	8	Service Indust	20	25	12
Earth/Clay/Stone	33	26	22	Earth/Clay/Stone	27	26	18
Financial Inst	20	27	16	Volunt./Church	25	27	24
Volunt./Church	27	28	26	Social Security	28	28	31
Fisheries	25	29	13	Train System	31	29	21
Mining	22	30	19	Restaurants	10	30	8
Social Security	30	31	23	Fisheries	30	31	33
Priv. Household	32	32	30	Construction	8	32	7
Restaurants	24	33	11	Priv. Household	32	33	30

Table 24: Alpha Regressions: Lasso Occupation Selection Results

Hourly Earnings (flwage)				Annual Earnings (fearn)			
Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order	Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Not Applicable	1	1	1	Not Applicable	1	1	1
Machine Fitter	9	2	19	Farm Manager	2	2	2
Farm Manager	3	3	2	Machine Fitter	16	3	7
Inspector	12	4	47	Mathematician	4	4	34
Mathematician	4	5	38	Inspector	10	5	48
Convey. Oper.	5	6	33	Architect/Engineer	9	6	38
SecurityServic	44	7	37	Transport.Oper	14	7	3
Architect/Engineer	7	8	12	Ofc.Worker Etc	13	8	27
Electr. Enginr	13	9	16	SecurityServic	21	9	24
Vendor	56	10	15	BusinessManagr	12	10	28
Janitor	18	11	34	Scientist	29	11	51
Ofc.Worker Etc	21	12	44	Janitor	28	12	16
Mailman	10	13	46	Eng.Tech.Expert	27	13	59
Labor/Craftsmn	27	14	7	Priv.Bus.Leadr	3	14	6
Priv.Bus.Leadr	2	15	3	Educator	17	15	29
Transport.Oper	35	16	4	Tool/Die Maker	73	16	43
BusinessManagr	11	17	49	Mailman	34	17	32
Food Producer	19	18	39	Insurance Rep.	5	18	8
Eng.Tech.Expert	15	19	35	Accountant	18	19	39
Pipe Fitter	48	20	20	Convey. Oper.	37	20	10
Printer Etc.	42	21	42	Soldier	45	21	35
Cook/Waiter	16	22	9	Lawyer	25	22	57
Insurance Rep.	6	23	5	Electr. Enginr	11	23	13
Accountant	32	24	22	Food Producer	23	24	53
Scientist	23	25	26	Administrator	60	25	55
Tool/Die Maker	25	26	58	Office Manager	30	26	52
Bricklay/Carpt	43	27	6	Pipe Fitter	15	27	9
Stat.Mach.Oper	37	28	65	Labor/Craftsmn	6	28	5
Educator	8	29	17	Vendor	32	29	12
Music/Perform	26	30	8	Lumbrman/Axman	8	30	20
Aero/MarineEngr	14	31	32	ChemicalWorker	65	31	64
Foundry Worker	71	32	63	ComputerOperat	39	32	68
ComputerOperat	33	33	78	Aero/MarineEngr	40	33	63
Chemist	40	34	48	RelatMedicalJob	20	34	11
Soldier	24	35	21	Bookkeepr/Cash	78	35	31
Administrator	67	36	55	Painter	38	36	17
Lumbrman/Axman	30	37	13	Bricklay/Carpt	7	37	4
Office Manager	38	38	53	Printer Etc.	57	38	30
Legislator	39	39	10	Tailor	77	39	45
Hair Stylist	36	40	11	Life/Phys. Sci.	26	40	58
Bookkeepr/Cash	47	41	36	Dr./Dentist/Vet	22	41	36
Dr./Dentist/Vet	22	42	29	Spinner/Weaver	53	42	69
Painter	45	43	27	Author	66	43	47
Stenographer	52	44	67	Hair Stylist	36	44	21
ChemicalWorker	68	45	68	Music/Perform	41	45	23
Spinner/Weaver	49	46	64	Buyer	33	46	66
RelatMedicalJob	63	47	28	Sculptr/Paintr	35	47	26
Lawyer	34	48	23	Cleric	48	48	40
Forestry Work	55	49	69	Fisher/Hunter	47	49	49
Broadcaster	77	50	43	Legislator	31	50	15
Cleric	75	51	62	Stat.Mach.Oper	51	51	61
Glazier	57	52	77	Domestic Help	62	52	60
Miner	50	53	57	Stenographer	54	53	71
Economist	66	54	52	Economist	72	54	62
Author	58	55	31	Cook/Waiter	24	55	14
Tailor	54	56	30	Tel. Operator	52	56	44
Fisher/Hunter	31	57	14	Rest./StoreMgr	46	57	25
Rest./StoreMgr	17	58	18	Transp. Attend	69	58	76
Prof. Athlete	46	59	41	Tech.Salespers	42	59	18
Tel. Operator	60	60	60	Service Worker	50	60	22
Dry-Cleaner	64	61	61	Chemist	56	61	42
Jewelry Maker	20	62	76	Dry-Cleaner	44	62	37
Agriculturladm	29	63	50	Cabinet Maker	49	63	54
Cabinet Maker	28	64	56	Salesperson	64	64	65
Salesperson	59	65	70	Glazier	58	65	74
Shoemaker	70	66	72	Agriculturladm	55	66	19
Buyer	61	67	59	Miner	76	67	56
Farm Hand	76	68	45	Forestry Work	59	68	70
Domestic Help	65	69	51	Conductor	75	69	75
Tech.Salespers	53	70	54	Shoemaker	61	70	50
Sculptr/Paintr	51	71	25	Broadcaster	74	71	67
Stone Cutter	73	72	71	Prof. Athlete	43	72	33
Service Worker	41	73	40	Foundry Worker	19	73	46
Life/Phys. Sci.	62	74	24	Jewelry Maker	67	74	73
Manufacturer	78	75	75	Farm Hand	63	75	41
Conductor	69	76	74	Manufacturer	71	76	77
HH Supervisor	72	77	73	HH Supervisor	70	77	72
Transp. Attend	74	78	66	Stone Cutter	68	78	78